

Chapter 2

Classification of Directional Patterns

After stabilizing 10000 trials of 301×301 sandpile grids (each with background 3, plus 3 extra grains at random sites), we examine the pattern formed by cells whose height differs from 3. Far from the center, along a fixed rational direction (n, m) , a periodic height pattern emerges. We classify these patterns for all directions with $0 \leq n, m \leq 7$.

2.1 General Description of Directional Patterns

When one examines a stabilised grid far from the centre, the set of sites whose height differs from the background 3 forms a collection of lines. Each line extends outward in a fixed direction and is composed of a minimal periodic unit of coloured cells that repeats many times along that direction, giving every observed direction its own characteristic stripe. All observed directions are rational—that is, the slope of every line is a rational number; no line with an irrational slope appeared in any of the 10 000 trials.

2.2 Classification Table for (n, m) , $0 \leq n, m \leq 7$

Table 2.1 shows the directional pattern for each pair (n, m) . Only seven distinct primitive (coprime) directions were observed across all 10000 trials. When a pair (n, m) reduces by $\gcd(n, m)$ to one of these seven primitive directions, the corresponding image is shown; otherwise the entry reads “not found.”

Table 2.1: Directional pattern classification for $0 \leq n, m \leq 7$. Rows index n , columns index m . Entries show the image of the reduced coprime direction.

$n \downarrow / m \rightarrow$	0	1	2	3	4	5	6	7
0								
1					not found	not found	not found	not found
2				not found		not found		not found
3			not found		not found	not found		not found
4		not found		not found		not found	not found	not found
5		not found	not found	not found	not found		not found	not found
6		not found			not found	not found		not found
7		not found	not found	not found	not found	not found	not found	

The seven observed directions are:

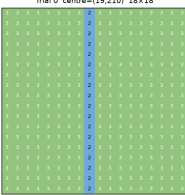
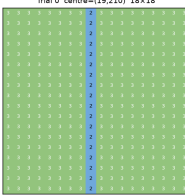
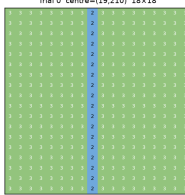
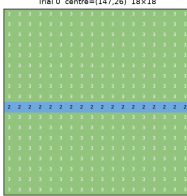
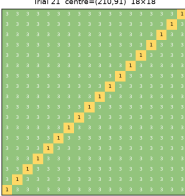
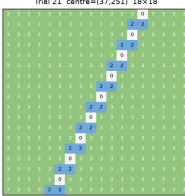
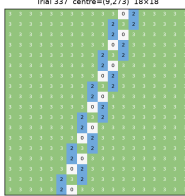
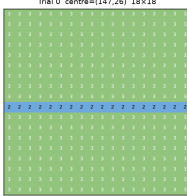
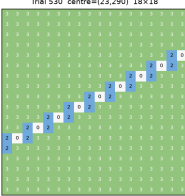
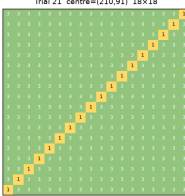
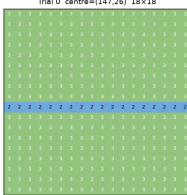
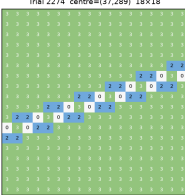
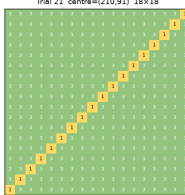
$$(0, 1), (1, 0), (1, 1), (1, 2), (1, 3), (2, 1), (3, 1).$$

Note that (n, m) and (m, n) yield symmetric patterns in general (e.g., $(1, 2)$ vs. $(2, 1)$ are symmetric). You can check the folder `images/` for the original images.

2.3 Enlarged Classification Table for (n, m) , $0 \leq n, m \leq 3$

Table 2.2 shows the same classification restricted to $0 \leq n, m \leq 3$. Because the table is smaller, each image is displayed at a larger size so the periodic structure of every pattern is easier to inspect. The mapping from (n, m) to its reduced coprime direction is the same as in Table 2.1; entries that do not correspond to any observed primitive direction are marked “not found.”

Table 2.2: Directional pattern classification for $0 \leq n, m \leq 3$ (enlarged). Rows index n , columns index m .

$n \downarrow / m \rightarrow$	0	1	2	3
0	—			
1				
2				not found
3			not found	

All seven primitive directions already appear within this $0 \leq n, m \leq 3$ range: $(0, 1)$, $(1, 0)$, $(1, 1)$, $(1, 2)$, $(1, 3)$, $(2, 1)$, and $(3, 1)$. The two remaining pairs, $(2, 3)$ and $(3, 2)$, are coprime but were not observed in any trial.